

PRACTICAL ON MAVEN

```
C:\Users\charl>mvn -version
Apache Maven 3.9.13 (39d686bd50d8e054301e3a68ad44781df6f80dda)
Maven home: C:\Users\charl\Downloads\apache-maven-3.9.13-bin\apache-maven-3.9.13
Java version: 17.0.18, vendor: Microsoft, runtime: c:\Program Files\Microsoft\jdk-17.0.18.8-hotspot
Default locale: en_US, platform encoding: Cp1252
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

1. Create Project

```
C:\Users\charl\myapp>mvn archetype:generate -DgroupId=com.example -DartifactId=myapp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> archetype:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< archetype:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- archetype:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO]
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: C:\Users\charl\myapp
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: myapp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: C:\Users\charl\myapp\myapp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 2.157 s
[INFO] Finished at: 2026-03-25T23:44:07+05:30
[INFO]
```

2. Go to Project Folder

```

C:\Users\charl\myapp>cd myapp

C:\Users\charl\myapp\myapp>dir
Volume in drive C is OS
Volume Serial Number is 5088-97B8

Directory of C:\Users\charl\myapp\myapp

03/25/2026  11:44 PM    <DIR>          .
03/25/2026  11:44 PM    <DIR>          ..
03/25/2026  11:44 PM                652 pom.xml
03/25/2026  11:44 PM    <DIR>          src
                    1 File(s)                652 bytes
                    3 Dir(s)  72,065,740,800 bytes free

```

3. Project Structure

- pom.xml → project config
- src/main/java → main code
- src/test/java → test code

4. Compile Code

5. Run Tests

6. Package Project

```

C:\Users\charl\myapp\myapp>mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:myapp >-----
[INFO] Building myapp 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.pom (7.9 kB
at 18 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/45/maven-plugins-45.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/45/maven-plugins-45.pom (8.4 kB at 155 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/maven-resources-plugin-3.4.0.jar (31 kB
at 440 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.pom (9.5 kB
at 184 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/47/maven-plugins-47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/47/maven-plugins-47.pom (7.2 kB at 153 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/47/maven-parent-47.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/47/maven-parent-47.pom (54 kB at 879 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/37/apache-37.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/37/apache-37.pom (26 kB at 582 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.14.2/junit-bom-5.14.2.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/junit/junit-bom/5.14.2/junit-bom-5.14.2.pom (5.7 kB at 132 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.15.0/maven-compiler-plugin-3.15.0.jar (84 kB
at 1.5 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.5.4/maven-surefire-plugin-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-surefire-plugin/3.5.4/maven-surefire-plugin-3.5.4.pom (4.9 kB a
t 104 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.5.4/surefire-3.5.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/surefire/surefire/3.5.4/surefire-3.5.4.pom (19 kB at 368 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/44/maven-parent-44.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/44/maven-parent-44.pom (52 kB at 985 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/34/apache-34.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/34/apache-34.pom (24 kB at 433 kB/s)

```

```

[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ myapp ---
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.2.0/file-management-3.2.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.2.0/file-management-3.2.0.pom (4.5 kB at 77 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/44/maven-shared-components-44.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/maven-shared-components/44/maven-shared-components-44.pom (3.8 kB at 72 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.19.0/commons-io-2.19.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.19.0/commons-io-2.19.0.pom (21 kB at 362 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/81/commons-parent-81.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/81/commons-parent-81.pom (78 kB at 1.3 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.pom (4.7 kB at 85 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.pom (32 kB at 604 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.pom (5.6 kB at 101 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.pom (3.5 kB at 60 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.pom
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.pom (2.0 kB at 39 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.21.0/commons-io-2.21.0.pom
Downloaded from central: https://repo.maven.apache.org/maven2/commons-io/commons-io/2.21.0/commons-io-2.21.0.pom (21 kB at 379 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/91/commons-parent-91.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-parent/91/commons-parent-91.pom (99 kB at 1.3 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.2.0/file-management-3.2.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/shared/file-management/3.2.0/file-management-3.2.0.jar (26 kB at 551 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.5/maven-archiver-3.6.5.jar (28 kB at 373 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.10.4/plexus-archiver-4.10.4.jar (227 kB at 2.5 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-io/3.6.0/plexus-io-3.6.0.jar (80 kB at 923 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-lang3/3.18.0/commons-lang3-3.18.0.jar (703 kB at 2.4 MB/s)
Downloading from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.jar
Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.7-6/zstd-jni-1.5.7-6.jar (7.4 MB at 8.4 MB/s)
[INFO] Building jar: C:\Users\charl\myapp\myapp\target\myapp-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO]
[INFO] Total time: 10.875 s
[INFO] Finished at: 2026-03-25T23:47:43+05:30
[INFO]

```

7. Run Program

```

C:\Users\charl\myapp\myapp>java -cp target/myapp-1.0-SNAPSHOT.jar com.example.App
Hello World!

C:\Users\charl\myapp\myapp>

```